

Jarvis: A Multimodal Visualization Tool for Bioinformatic Data

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Motivations

- Curated biological datasets can be challenging to navigate and explore;
 - e.g., size of protein-protein interaction networks.
- Information visualization is a valuable navigation and understanding tool
 - provided the visualization has the ability to manipulate large quantities of data;
 - including interactions between different visualization techniques (Tao et al., 2004; O'Halloran et al., 2018).

Motivations

- Bioinformatic naming schemes and ontologies discourage speech-based interactions;
 - e.g., difficulty pronouncing or resolving entities;
 - “angiotensin-converting enzyme 2” → **ACE2** (“ace-two”)
- Other modalities may be better at grounding certain types of information:
 - e.g., deictic gesture to locations.

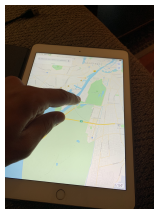


Figure: “We should be *here*.”

Motivations

- Multimodal methods for manipulating data can be helpful, e.g.,
 - Voice commands with demonstratives such as “this” and “that”;
 - Haptic interface can specify intended targets.
- **Jarvis** combines speech and haptic controls to encourage robust and flexible Q+A interactions;
 - Test use case: Biocuration data;
 - More generally, data exploration with multimodal queries, e.g.,
 - language input, pointing, swiping, tapping, etc.
- We hope to enhance the navigability and potential for discovery of results returned through complex queries.

Prior Work

- Tablets/smartphones made haptic interfaces commonplace;
- Improved speech recognition did the same for voice commands.
- Integrative crossmodal interfaces (e.g., localizing to maps) (Johnston, 2009; Selfridge and Johnston, 2015; Johnston, 2019).
- Analysis of both biological and biomedical literature (e.g., Kim, 2017; Lee et al., 2020).
- Language interaction over biocuration datasets (Friedman et al., 2017; Gyori et al., 2017; McDonald et al., 2016; Todorov et al., 2019 - <http://pathwaymap.indra.bio>).

Prior Work

- Visualization of dense data for bioinformatics via clustergrams (Schonlau, 2002).
- Dimensionality reduction through manifold approximation and principal component analysis (Tao et al., 2004; Clark and Ma'ayan, 2011; McInnes and Healy, 2018).
- Other visualization tools e.g., Cytoscape, ProViz (Shannon et al., 2003; Jehl et al., 2016).
- Jarvis brings together multiple modalities, background data from bioinformatic literature, and the aforementioned visualization techniques.

Architecture

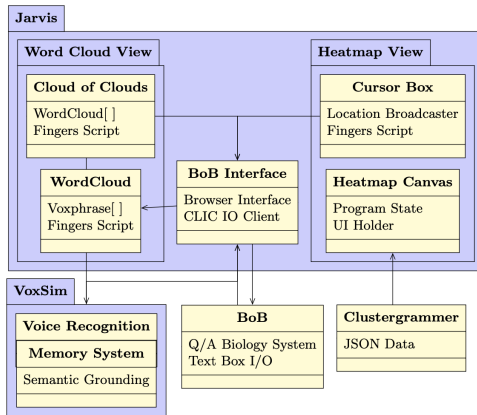
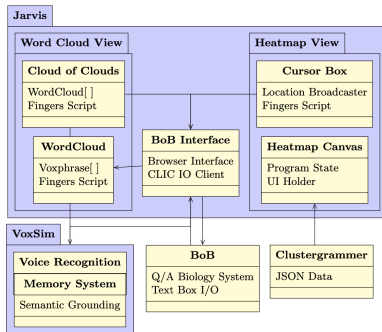


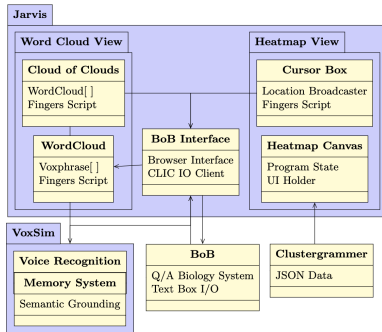
Figure: The main components of Jarvis

Architecture



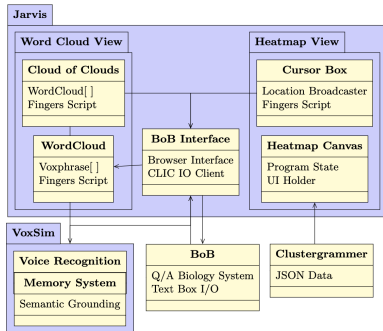
- **VoxSim**: semantic visualization engine that facilitates manipulating visualized objects.
- **BoB**: dialogue system for biological Q+A.
- **Clustergrammer**: input data about e.g., gene-protein interactions in the form of clustergrams.
- **Heatmap Canvas**: visualizes the heatmap from Clustergrammer data.

Architecture



- User can zoom on a desired location.
- Data within the bounds is saved and sent to BoB.
- **Voice Recognition** recognizes the query and semantically grounds entities in it to entities in the world/heatmap data.
- BoB processes query and returns subset of the genes it receive.

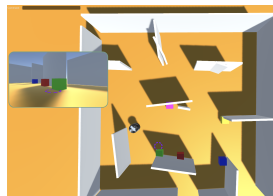
Architecture



- **Word Cloud View** received both the original set and subset (Cloud 1: visualized full set; Cloud 2: visualized subset).
- Individual proteins can be manipulated in the Word Cloud View
- Subsequent questions can be asked about those proteins.

VoxSim

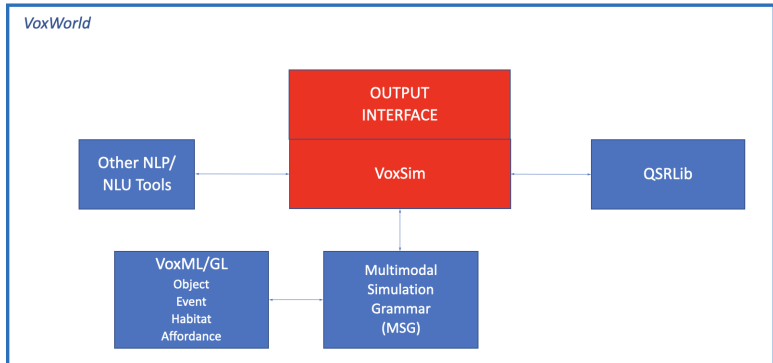
- Unity game engine-based event simulator used to develop intelligent agent behaviors;
 - Typical use-case: interactive embodied agents.



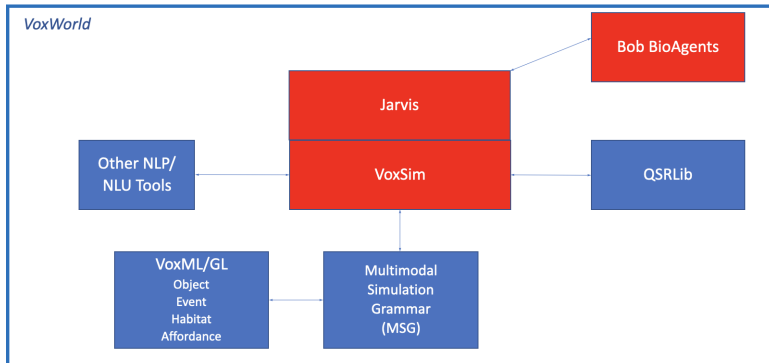
VoxSim

- VoxSim is built on the modeling language VoxML (Pustejovsky and Krishnaswamy, 2016);
 - VoxML encodes the semantics of visualized lexical items (*voxemes*)
 - Allows visualized objects to be manipulated in 2D and 3D space
- Jarvis exploits the abstract properties of voxemes to render elements from the underlying dataset as manipulable objects to facilitate data exploration.

VoxSim



VoxSim



Usage

- Objects are made interactable through VoxSim and the *Fingers* Unity asset, which parallels native iOS gestures.
- Speech recognition via Google SR.
 - VoxSim supports arbitrary 3rd-party endpoints.
- Heatmap data via Clustergrammer;
 - Groups hierarchically-clustered heatmaps from gene expression data and saves them to JSON.
 - Heatmap visualization algorithm is also adapted from Clustergrammer.

Haptic Control

- Use of gestures depends on the visualized context and technique (e.g., heatmap or word cloud).
- Some gestures may mean different things in different contexts.
- The gesture *inhabits* the visual context:
 - $H_{[1]}$ = in heatmap; $H_{[2]}$ = in word cloud.

Gesture	Heatmap Interpretation	Word Cloud Interpretation
Tap	—	Semantically ground word
Swipe	Swap View	Swap View
Pan	Move selection box	—
Scale	Resize selection box	Zoom camera
Rotate	—	Rotate Word Cloud
Long Press	—	Semantically ground cloud

Table: Gestures available in Jarvis

Multimodal Integration

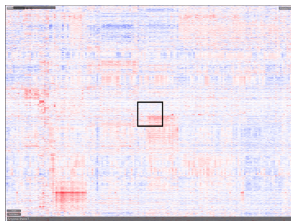


Figure: A heatmap of gene expression vs. tissue samples with UI overlaid

- This heatmap represents gene expression over tumor samples (rows: genes; columns: source tissue).
- Haptics on tablet drag and resize the selector box;
- Voice (e.g., “zoom in here”) zooms in, grounds the selected area to the data.

Multimodal Integration

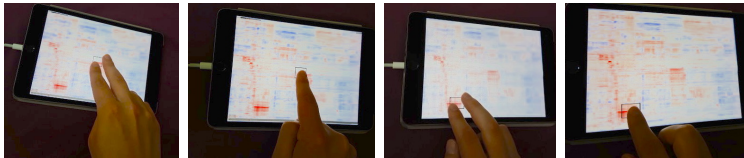


Figure: Interacting with Jarvis interface

- With a query, e.g., “Which of **these** are transcription factors?” the system reads the currently selected genes and passes the list and query to BoB;
- The resultant subset is passed back to Jarvis for visualization.

Multimodal Integration



Figure: Clouds of proteins names for manipulation

- The two lists are then visualized as word clouds;
 - Each word in the cloud is individually interactable.
- Each word's position in the clouds is determined by factors in the underlying data, such as frequency of occurrence or similarity to another data point.

Multimodal Integration

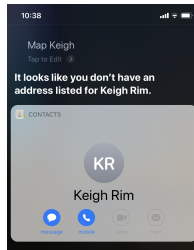


Figure: Clouds of proteins names for manipulation

- The clouds respond to haptic input as well;
 - A subset of the cloud, e.g. which proteins are transcription factors, may be pulled out and manipulated independently.
- The cloud can be reordered to correspond to relationships with a selected word.

Multimodal Integration

- The nature of bioinformatic data, particularly protein and gene names, poses a problem for purely speech-based systems;
- Gene names may be initialisms or difficult to pronounce;
 - e.g., “MAPK” has a conventionalized pronunciation (“map-kay”).
 - Say this to a smart phone:



Multimodal Integration

- Navigating bio domain via speech alone is difficult.
- Providing text input helps but is time-consuming and doesn't solve the problem if user doesn't know how to phrase requests.
- Large bioinformatic datasets are typically presented visually (heatmaps, graphs, word clouds);
 - Not necessarily easy to interact with these linguistically.
- Haptics to indicate regions or entities of interest allows simpler language ("this," "that," "these," etc.);
 - Obviates the need to pronounce or spell out entity references.
- Makes navigation more tractable for a large dataset.
- Allows for less discursive language to ask the same question.

Evaluation

- Late-breaking paper: evaluation still planned.
- Goal of Jarvis in this use case is well-defined: to enable biologists to accomplish novel discoveries in large datasets;
 - We can evaluate the usability of the system in accomplishing this task.
- We can also evaluate interactive capabilities for data exploration over arbitrary large datasets;
- Strictly biological data is not a requirement for useful multimodal exploration, it is simply an illustrative use case.

Interactive Usability

- We propose a multimodal evaluation method based on Krishnaswamy and Pustejovsky, 2018.
- An interaction consists of timestamped “moves” coded by participant and modality (*S* for speech, *G* for gesture, *A* for action).

1	JARVIS _A	CREATE_HEATMAP(data[])	0.000000
2	USER _G	PAN_TO (<.14674;.24371>)	1.145281
3	USER _S	"Which of these are transcription factors?"	2.452981
4	BoB _S	"I am having trouble, possibly because I don't know what 'these' refers to."	5.803915
5	BoB _S	"I don't know what genes you mean."	7.818170
6	USER _S	"Create the gene set."	8.642095
7	BoB _S	"Okay."	10.041973
8	BoB _S	"I created the gene-set selection with 7 items."	12.803915
9	BoB _S	"What would you like to do next?"	14.500183
10	USER _S	"Which of these are transcription factors?"	15.661427
11	JARVIS _A	CREATE_WORDCLOUD(geneset[])	18.891054
12	JARVIS _A	CREATE_WORDCLOUD(subset[])	18.891054
13	BoB _S	"Of those 7 genes, PLAGL1 is a transcription factor."	18.891054

Table: Sample interaction log.

Interactive Usability

- Moves 0-1: Jarvis presents heatmap, facilitating the user selecting a region by haptic panning;
- We can quantify time between data presentation and user interacting with it.
 - A long delay may signal confusion on the part of the user, e.g., in how to use the system and/or what to do with data presentation.
- Dialogue breaks down at move 3: user says something that BoB does not understand, so BoB gives the user a reason why.
- User responds with new instruction that grounds the demonstrative “these” to a particular set.
- Jarvis facilitated repair at move 6.

Fidelity of Data Transfer

- **Speech Recognition:** Have users execute a scripted dialogue that provides a known ground truth, then compare the recognized input to that reference using, e.g., BLEU score (Papineni et al., 2002).
- **Semantic Grounding:** Look at “blocks” in the log bounded by moves that negate a prior move and redirect the interaction to new focus objects or actions.
 - Information is being correctly grounded within an block.
 - Longer blocks means better continuous grounding performance.
 - Can assess grounding through language and grounding through haptics.

Fidelity of Data Transfer

- **Visualization Accuracy:**

- Match list of genes passed to BoB to list of visualized gene names.
- Have user select same region multiple times through mouse or haptics and calculating overlap between successive selections.
- Correlate region selection to underlying data by calculating overlap in retrieved data over multiple selections.

Conclusion

- Jarvis, combines speech and haptic control with a robust biocuration dialogue system;
- These features encourage smooth interactions over complex data.
- The underlying mechanism that enables the integration of the two distinct modalities is the transformation of data into a manipulable object.
- This allows domain specialists to navigate through the data using multiple grounding techniques.
- The same underlying interface is adaptable to multiple datasets potentially in multiple domains.

Future Work

- 3D visualizations of the heatmap view add an additional dimension
 - Useful for representing time sequence data, e.g., single-cell or cell-cycle proliferation data.
- More extended commands for data manipulation.
 - e.g., double- or triple-taps to highlight relationships or rearrange elements.
- More in-depth gene-to-gene relationships in the word cloud visualization.
 - e.g., grouping genes by known clusters, similar protein encoding, or generalized functions.
- Better integration between Fingers and VoxSim.
 - Use the VoxSim contextual memory to track objects the user has previously interacted with.
- “Save state” in a visualization and revisit it (e.g., the “swipe” gesture to swap views).

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